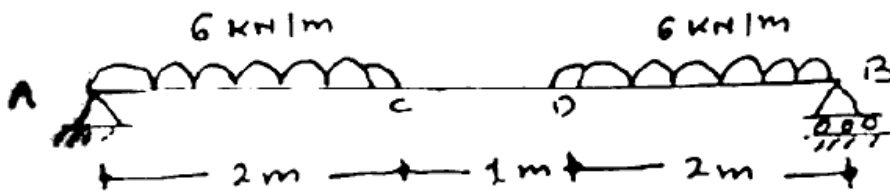
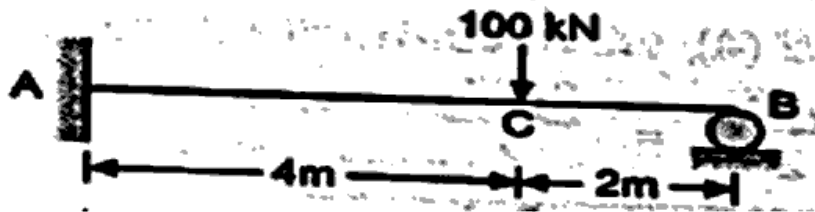
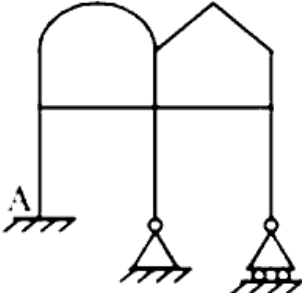
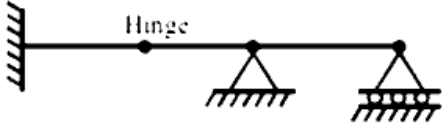
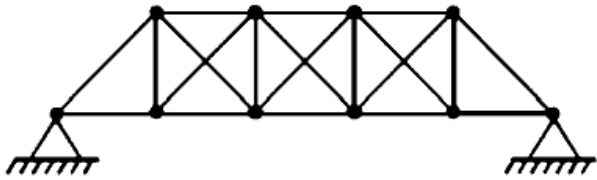


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular & Supplementary Summer 2024 Course: B. Tech. Branch : Civil Engineering Semester :IV Subject Code & Name : BTCVC403 Structural Mechanics - I Max Marks: 60 Date: 18/06/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q.1	Solve Any Two of the following.		12
A)	List out the different methods to find slope and deflection and explain them in detail?	CO2	6
B)	Derive an expression for slope and deflection for simply supported beam with a UDL throughout span using double integration method?	CO2	6
C)	Compute the max deflection of beam shown in fig below. Take $E = 2 \times 10^5 \text{ Mpa}$ and $I = 3 \times 10^7 \text{ mm}^4$. Use conjugate beam method. 	CO2	6
Q.2	Solve Any Two of the following.		12
A)	Using strain energy method and determine the deflection at free end of cantilever of length L subjected to a concentrated load 'P' at the free end.	CO2	6
B)	A simply supported beam of span 9m carries two point loads, each 50KN at 3m from each support. Using castigliano's first theorem. Find the deflection at mid span. Take $EI = 25 \times 10^3 \text{ KN-m}^2$.	CO2	6
C)	Determine reaction at B for the propped cantilever AB as shown in fig below. Assuming constant flexural rigidity (EI)	CO2	6

			
Q.3	Solve Any Two of the following.		12
A)	A fixed beam of span 'L' subjected to a UDL of W KN/m for a span of 'a' from left support. Determine the end moments developed.	CO2	6
B)	Determine the indeterminacy for the following. i) Determine the degree of freedom of the following frame from fig 1. ii) The degree of indeterminacy of the beam in fig 2 is. iii) The total degree of indeterminacy (both internal and external) for the bridge truss shown in the given fig 3 is  Fig 1  Fig 2  Fig 3	CO1, CO3	6
C)	A continuous beam ABCD is simply supported over three spans, such that AB=8m, BC=12m and CD=5m. It carries a UDL of 4KN/m in span AB, 3KN/m in span BC and 6KN/m in span CD. Find the support moments over supports B and C. Assuming constant EI draw shear force and bending moment diagram.	CO2, CO3	6
Q.4	Solve Any Two of the following.		12
A)	Define the following terms: a) Carry over factor b) Distribution factor c) Carry over moment d) Relative stiffness.	CO2	6
B)	Analyze the continuous beam as shown below and draw BMD using moment distribution method	CO2	6

C)	Analyze the frame shown below for moments at the ends of members. Draw BMD. EI is same for all the members. Use Moment distribution method.	CO2	6
Q.5 Solve Any Two of the following.			12
A)	Determine the support moments for the continuous beam shown below. Draw BMD. Use slope deflection method.	CO2	6
B)	Determine the support moments for the continuous beam shown below. Draw BMD. Use slope deflection method	CO2	6
C)	Analyse the frame given in figure by moment distribution method and draw the B.M.D&S.F.D	CO2	6
*** End ***			